

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Video Streaming Quality Adjustment System

A streaming system processes frames using the recurrence $T(n) = 2T(n/2) + 1$. Apply the Master Theorem to determine the time complexity. Also analyze the space complexity

Q2. Medical Diagnosis Support System

A medical diagnosis system analyzes patient data using the recurrence $T(n) = 2T(n/4) + n$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how this affects performance for large hospital datasets. Interpret efficiency using asymptotic notation.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Video Streaming Quality Adjustment System

A streaming system processes frames using the recurrence $T(n) = 2T(n/2) + 1$. Apply the Master Theorem to determine the time complexity. Also analyze the space complexity.

Master Theorem

The Master Theorem is used to determine the time complexity of recurrence relations of the form:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

where:

- $a \geq 1$ = number of recursive subproblems
- $b > 1$ = factor by which the problem size is divided
- $f(n)$ = work done outside the recursive calls

Let

$$c = \log_b a$$

Then the Master Theorem has three cases:

Case 1

If

$$f(n) = O(n^{c-\varepsilon})$$

for some constant $\varepsilon > 0$, then

$$\boxed{T(n) = \Theta(n^c)}$$

Case 2

If

$$f(n) = \Theta(n^c \log^k n)$$

for some constant $k \geq 0$, then

$$\boxed{T(n) = \Theta(n^c \log^{k+1} n)}$$

Case 3

If

$$f(n) = \Omega(n^{c+\varepsilon})$$

for some constant $\varepsilon > 0$, and the regularity condition

$$af\left(\frac{n}{b}\right) \leq cf(n)$$

for some constant $c < 1$ holds, then

$$\boxed{T(n) = \Theta(f(n))}$$

This is the standard statement of the Master Theorem used to solve divide-and-conquer recurrence relations.

Problem

A streaming system processes frames using the recurrence:

$$T(n) = 2T\left(\frac{n}{2}\right) + 1$$

Determine the time complexity using the Master Theorem and analyze the space complexity.

Step 1: Identify Master Theorem Parameters

The standard recurrence is:

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

Here,

- $a = 2$
- $b = 2$
- $f(n) = 1 = \Theta(1)$

Step 2: Calculate $n^{\log_b a}$

$$\log_2 2 = 1$$

Therefore,

$$n^{\log_b a} = n^1 = n$$

Step 3: Compare $f(n)$

$$f(n) = 1 = \Theta(1)$$

Since

$$1 = O(n^{1-\varepsilon})$$

where $\varepsilon = 1$,

this satisfies Case 1 of the Master Theorem.

Step 4: Apply Master Theorem

Case 1 gives

$$T(n) = \Theta(n)$$

Time Complexity

$$\boxed{\Theta(n)}$$

Space Complexity

The recursion depth is: $\log_2 n$

Each recursive call requires constant stack space.

Therefore,

$$\boxed{\Theta(\log n)}$$

Performance Analysis

- Every frame is processed once.
- Work at each level is constant.
- Recursive levels are logarithmic.
- Overall execution grows linearly with the number of video frames.

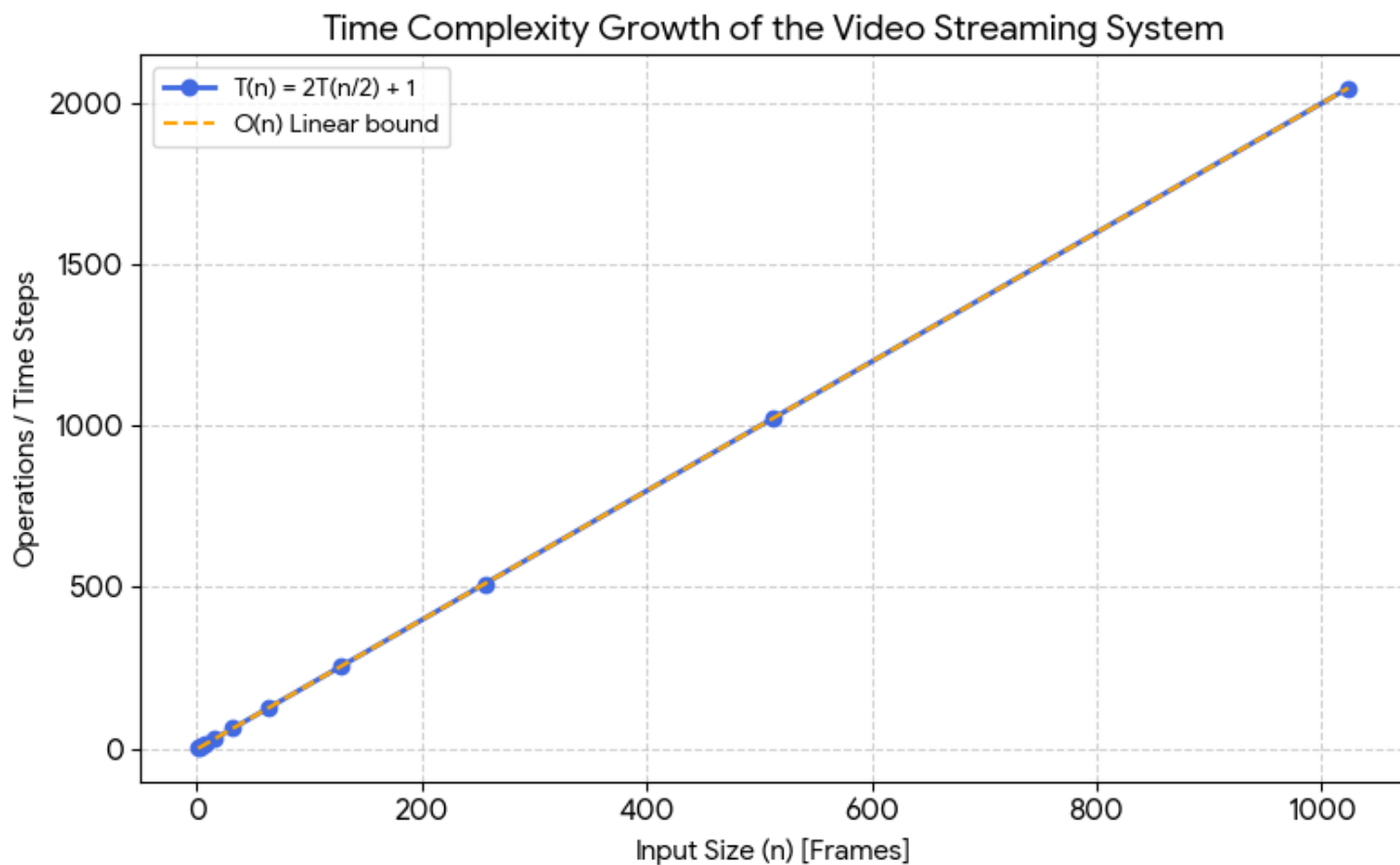
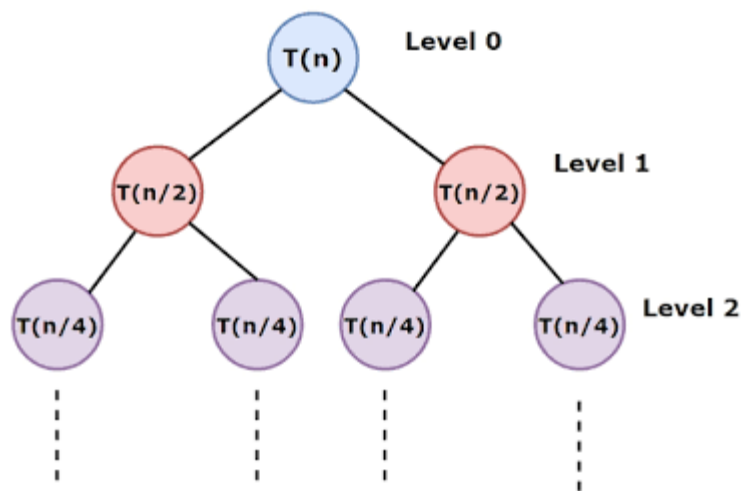
Final Answer

Time Complexity

$$\boxed{\Theta(n)}$$

Space Complexity

$$\boxed{\Theta(\log n)}$$



Q2. Medical Diagnosis Support System

A medical diagnosis system analyzes patient data using the recurrence $T(n) = 2T(n/4) + n$. Apply the Master Theorem to determine the time complexity. Identify parameters and justify the case selection. Analyze how this affects performance for large hospital datasets. Interpret efficiency using asymptotic notation.

What is Asymptotic Notation?

Asymptotic notation is a mathematical method used to describe the growth rate of an algorithm's time or space complexity as the input size n becomes very large. It helps compare the efficiency of algorithms independent of hardware or programming language.

The three main asymptotic notations are:

- **Big-O (O):** Represents the **upper bound** (worst-case time complexity).
- **Big-Omega (Ω):** Represents the **lower bound** (best-case time complexity).
- **Big-Theta (Θ):** Represents the **tight bound** (exact growth rate).

Given Recurrence

$$T(n) = 2T\left(\frac{n}{4}\right) + n$$

We solve it using the **Master Theorem**.

Step 1: State the Master Theorem

A recurrence of the form

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

where:

- a = number of recursive subproblems
- b = factor by which the problem size is divided
- $f(n)$ = work done outside the recursive calls

Step 2: Identify the Parameters

Comparing

$$T(n) = 2T\left(\frac{n}{4}\right) + n$$

with

$$T(n) = aT\left(\frac{n}{b}\right) + f(n)$$

we get:

- $a = 2$
- $b = 4$

- $f(n) = n$

Step 3: Compute $n^{\log_b a}$

$$\log_4 2 = \frac{1}{2}$$

Therefore,

$$n^{\log_4 2} = n^{1/2} = \sqrt{n}$$

Step 4: Compare $f(n)$

$$f(n) = n$$

Since

$$n = \Omega(n^{1/2+\varepsilon})$$

(for example, $\varepsilon = \frac{1}{2}$), $f(n)$ grows faster than $n^{\log_b a}$.

Also, the regularity condition holds:

$$2f\left(\frac{n}{4}\right) = 2\left(\frac{n}{4}\right) = \frac{n}{2} \leq cn$$

where $c = \frac{1}{2} < 1$.

Hence, **Case 3** of the Master Theorem applies.

Step 5: Apply the Master Theorem

According to **Case 3**,

$$T(n) = \Theta(f(n))$$

Since

$$f(n) = n,$$

the solution is

$$\boxed{T(n) = \Theta(n)}$$

Time Complexity

$$\boxed{\Theta(n)}$$

The running time increases linearly with the number of patient records.

Space Complexity

The recursion depth is

$$\log_4 n$$

Each recursive call uses constant stack space.

Therefore,

$$\Theta(\log n)$$

Performance for Large Hospital Datasets

- The patient data is divided into **2 smaller subproblems**, each of size $n/4$.
- The system performs **linear work (n)** to process patient information at each stage.
- As the number of patient records increases, the execution time increases **linearly**.
- Therefore, the algorithm scales efficiently and is suitable for handling large hospital databases.

Interpretation Using Asymptotic Notation

- **Big-O:** $O(n)$ – Worst-case running time.
- **Big-Theta:** $\Theta(n)$ – Exact running time.
- **Big-Omega:** $\Omega(n)$ – Best-case lower bound.

Thus,

$$O(n) = \Theta(n) = \Omega(n)$$

Final Answer

- **Recurrence:** $T(n) = 2T(n/4) + n$
- **Parameters:** $a = 2, b = 4, f(n) = n$
- **Master Theorem Case:** Case 3
- **Time Complexity:** $\Theta(n)$
- **Space Complexity:** $\Theta(\log n)$
- **Performance:** Linear growth makes the algorithm efficient for large hospital datasets.
- **Asymptotic Notation:** $O(n), \Theta(n), \Omega(n)$

